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Brain Tumor Segmentation Based on the Fusion of Deep Semantics and Edge Information in Multimodal MRI

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Abstract

Brain tumor segmentation in multimodal MRI has great significance in clinical diagnosis and treatment. The utilization of multimodal information plays a crucial role in brain tumor segmentation. However, most existing methods focus on the extraction and selection of deep semantic features, while ignoring some features with specific meaning and importance to the segmentation problem. In this paper, we propose a brain tumor segmentation method based on the fusion of deep semantics and edge information in multimodal MRI, aiming to achieve a more sufficient utilization of multimodal information for accurate segmentation. The proposed method mainly consists of a semantic segmentation module, an edge detection module and a feature fusion module. In the semantic segmentation module, the Swin Transformer is adopted to extract semantic features and a shifted patch tokenization strategy is introduced for better training. The edge detection module is designed based on convolutional neural networks (CNNs) and an edge spatial attention block (ESAB) is presented for feature enhancement. The feature fusion module aims to fuse the extracted semantic and edge features, and we design a multi-feature inference block (MFIB) based on graph convolution to perform feature reasoning and information dissemination for effective feature fusion. The proposed method is validated on the popular BraTS benchmarks. The experimental results verify that the proposed method outperforms a number of state-of-the-art brain tumor segmentation methods. **The source code of the proposed method is available at <https://github.com/HXY-99/brats>.**

Keywords: Brain tumor segmentation, Transformer, convolutional neural networks, edge feature, feature fusion

1. Introduction

Medical image segmentation is an important topic in the community of medical image processing. Among them, brain tumor segmentation aims to localize multiple types of tumor regions from images, which is of

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great significance to clinical practice [1]. Owing to the good capacity in providing high-resolution anatomic structures for soft-tissues, magnetic resonance imaging (MRI) is mostly utilized in the diagnosis and treatment of brain tumor diseases. To obtain comprehensive information for accurate segmentation, multimodal MRI scans with different imaging parameters are usually required in brain tumor segmentation. Commonly-used modalities include fluid attenuation inversion recovery (FLAIR), T1-weighted (T1), contrast enhanced T1-weighted (T1ce) and T2-weighted (T2). Images of different modalities capture different pathological information and they can complement each other effectively [2], which plays a crucial role in segmenting multiple types of brain tumor regions such as edema (ED), necrosis and non-enhancing tumor (NCR/NET), and enhancing tumor (ET). An example of multimodal MRI for brain tumor segmentation is shown in Fig. 1. For simplicity, only a slice is selected from the entire scan. Fig. 1(a) shows the ground truth (GT) segmentation label provided by domain experts. The green, yellow and red indicate ED, ET and NCR/NET regions, respectively. From Fig. 1(b)-(e), it can be seen that the characteristics of different modalities vary significantly. For example, the FLAIR modality can well capture the ED regions with distinct edges or boundaries between the tumor and normal tissues, while the T1ce modality is more effective in detecting the tumor core (i.e., the union of ET and NCR/NET) with high contrast [3].

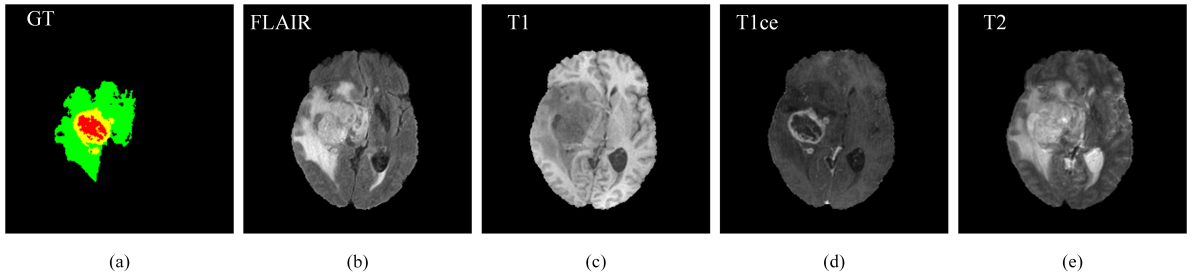


Figure 1: An example of multimodal MRI for brain tumor segmentation. (a) The ground truth (GT) segmentation label provided by domain experts (the green, yellow and red represent edema (ED), enhancing tumor (ET), and necrosis and non-enhancing tumor (NCR/NET), respectively). (b) The FLAIR modality. (c) The T1 modality. (d) The T1ce modality. (e) The T2 modality.

A variety of brain tumor segmentation approaches have been presented in the literature. In recent years, deep learning-based methods have emerged as the mainstream in this field [4]. The most popular way is to adopt semantic segmentation-oriented convolutional neural networks (CNNs) such as fully convolutional networks (FCNs) [5], U-Net [6] and V-Net [7] to segment brain tumors. The CNN-based models can well capture local features in 2D or 3D spaces. However, CNNs are limited by the receptive field of convolutions, leading to difficulty in characterizing the global dependencies of features, which is essentially an important clue in semantic segmentation. To address this issue, Transformer-based models [8, 9] have been introduced into the study of brain tumor segmentation [10]. By establishing the connection between feature base units (i.e., tokens) and adopting a self-attention mechanism, the Transformer-based models demonstrate better capacity in capturing global contextual information. However, the input size of the standard vision

Transformer (ViT) [9] is fixed, which causes the problem of excessive computational cost for semantic segmentation that requires pixel-wise dense prediction. By constructing hierarchical structure to obtain feature maps like CNNs, the Swin Transformer [11] well solves this problem and achieves efficient dense prediction, which exhibits clear advantages for semantic segmentation problems. Nevertheless, the Swin Transformer still suffers from the problem of low locality inductive bias, which means that it usually requires a very large amount of training data to obtain satisfactory visual representation [12]. Since the dataset size of most medical image analysis problems is typically very small, a pre-trained model is generally needed when utilizing ViT and its variants including Swin Transformer. However, an appropriate pre-trained model is not always available in practice and performing pre-training by oneself is not an easy task.

In most existing brain tumor segmentation methods, the multimodal MRI scans are simply stacked as the model input for semantic segmentation, which may cause the insufficient utilization of multimodal information [13, 14]. In fact, the role of some modalities tends to be more significant in the segmentation task, as they contain more distinctive information. For instance, the FLAIR and T1ce modalities can effectively capture the edges of multiple types of tumor regions such as ED and ET. The edge information is actually very important with regard to brain tumor segmentation, as it not only helps to achieve better localization for the tumors, but also benefits the boundary quality (e.g., sharpness and accuracy) of segmented regions [15, 16]. It is believed that the edge information could be a good complementary to the deep semantic information. It is worth noting that although some methods [2, 17, 18] consider the complementary features of multimodal inputs and explore multimodal fusion accordingly, they mainly focus on the extraction and fusion of deep semantic features without considering the importance of the specific edge features for segmentation.

In this paper, we propose a brain tumor segmentation method based on the fusion of deep semantics and edge information in multimodal MRI, aiming to achieve a more sufficient utilization of multimodal information for accurate segmentation. Specifically, the proposed segmentation framework consists of three main modules: semantic segmentation, edge detection and feature fusion. The semantic segmentation module adopts the Swin Transformer as the backbone due to its advantages mentioned above. Moreover, we introduce a shifted patch tokenization strategy [12] into the Swin Transformer to increase its locality induction bias, so as to achieve easier training for small-size datasets. The edge detection module is designed based on CNNs to extract edge features from the FLAIR and T1ce modalities by considering their characteristics. The feature fusion module is designed to fuse semantic features and edge features extracted from MRI of different modalities. This module adopts graph convolution to structure different areas into different vertices and collects similar semantic features and edge features under the same vertex. It realizes the reasoning and dissemination of information between semantic features and edge features, leading to the improvement of feature fusion effect. Through the above designs, the proposed segmentation framework can effectively extract and fuse deep semantic features and edge features in multimodal MRI, and experimental results on BraTS benchmarks in 2018, 2019 and 2020 demonstrate its superior performance in brain tumor

segmentation.

In summary, the contributions of this paper are four-fold.

1. The primary contribution of this paper is that we propose a deep learning-based brain tumor segmentation method that simultaneously utilizes deep semantic features and specific edge features. To the best of our knowledge, this manner is different from existing works that mostly focus on the extraction and fusion of semantic features from multimodal MRI. To achieve this target, three modules including a semantic segmentation module, an edge detection module, and a feature fusion module are designed, leading to the following three technical contributions.
2. We present a Swin Transformer-based semantic segmentation module to extract semantic features for brain tumor segmentation. In particular, to address the problem caused by the lack of locality inductive bias, we introduce a shifted patch tokenization strategy into the Swin Transformer, leading to easier training for small-size datasets.
3. We present a CNN-based edge detection module to extract edge features from the FLAIR and T1ce modalities. In this module, an edge spatial attention block (ESAB) using Sobel operator is designed to enhance edge features, which are extracted in a progressive manner.
4. We present a feature fusion module to fuse the extracted deep semantic features and edge features. Specifically, a multi-feature inference block (MFIB) based on graph convolution is designed to achieve effective feature fusion.

2. Related Work

Various methods for brain tumor segmentation have been proposed in recent years. These methods can be broadly classified into two categories: the generative model-based methods and the discriminative model-based methods [19]. The generative model-based methods focus on the appearance characteristics of tumorous and healthy tissues, thus requiring related domain-specific prior information, which is usually obtained through probabilistic image atlases. Menze et al. [20] augmented a probabilistic atlas of healthy tissue priors with a latent atlas of the lesion and derive the estimation algorithm to extract tumor boundaries and the latent atlas from the image data. Heinrich et al. [21] employed discrete optimization and self-similarity for multimodal medical image segmentation under discrete medical image registration framework. The discriminative model-based methods regard tumor segmentation as a classification problem to determine the property of voxels. Owing to the rapid development of machine learning techniques, the discriminative model-based methods have gradually become the main trend in this field. Early methods in this category

mainly rely on hand-crafted features such as local histograms [22] and texture features [23], and then employ
95 discriminative models such as decision tree [24] and conditional random field [25] for classification.

In the past few years, deep learning has rapidly become the mainstream in the study of brain tumor
segmentation. Some early approaches adopt a patch-based classification strategy and design CNNs to
predict the class of the center voxel of a 2D or 3D patch [19, 26]. However, it is difficult for such patch-
based methods to fully consider the correlation among neighboring patches within a relatively large range.
100 To address this issue, end-to-end semantic segmentation models such as U-Net [6], attention U-Net [27]
and U-Net++ [28] have become more popular in brain tumor segmentation. Myronenko [3] proposed a
segmentation network that adds a variational auto-encoder branch to reconstruct the input image for more
effective feature learning. Liu et al. [29] introduced pixel-level image fusion as an auxiliary task to regularize
feature learning and presented a multi-task model for brain tumor segmentation. Isensee et al. [30] proposed
105 an adaptive framework based on 2D U-Net, 3D U-Net and U-Net Cascade. The framework automatically
adjusts all hyperparameters without human intervention.

Although the CNN-based methods have achieved great success in brain tumor segmentation, it is known
that CNNs suffer from the limitation of capturing global contextual information, which is a crucial clue
for semantic segmentation. To solve this problem, the Transformer-based methods have gained increasing
attention in the field of medical image segmentation with some representative models being proposed, such as
110 the TransUNet [31] that combines Transformer and U-Net, the MedT [32] that presents gated axial-attention
for segmentation. In the study of brain tumor segmentation, the TransBTS proposed by Wang et al. [10] is
the first work that uses Transformer for segmentation and achieves good performance. The above methods
are based on the standard ViT [9], in which the input size should be fixed, leading to high computational
115 complexity for dense prediction problems such as semantic segmentation. The Swin Transformer [11], which
adopts hierarchical structure to obtain feature maps like CNNs, can effectively alleviate this problem. This
improvement greatly enhance the potential of Transformer models for semantic segmentation, and motivates
us to adopt the Swin Transformer for brain tumor segmentation in this work. Nevertheless, similar to the
standard ViT, the Swin Transformer still suffers from a defect, i.e., the lack of locality inductive bias [12].
120 As a consequence, it is pretty hard to utilize the Swin Transformer for small-size datasets without pre-
training, leading to some inconvenience when applying it to medical image analysis tasks including brain
tumor segmentation, since an appropriate pre-trained model is not always available in practice. To tackle
this problem, we introduce a shifted patch tokenization strategy [12] into the Swin Transformer for brain
tumor segmentation, so that the model can be trained from scratch.

125 In order to obtain more accurate segmentation results, the use of multimodal MRI data has become an
interesting topic in brain tumor segmentation. Most existing methods simply adopt an multi-channel input
by stacking multimodal MRI scans and don't fully consider the difference in terms of their importance to
brain tumor segmentation. Pereira et al. [33] designed a convolutional network for automatic brain tumor

segmentation using a four-channel format for multimodal images. Wang et al. [34] achieved the segmentation of different brain tumor regions by constructing three cascaded networks. In order to use multimodal information more effectively, some feature fusion and selection approaches have appeared based on specific architectures such as attention mechanism. Dolz et al. [35] extended dense connections to multimodal image segmentation based on DenseNets. Each modality to be input into the network individually is considered as a branch and dense connections between each branch are used to fuse features from different modalities. Liu et al. [36] proposed an attention-based modality selection feature fusion module for multimodal feature refinement to address the difference among multiple modalities for a given segmentation target. Zhang et al. [37] used FCN to extract features from images of different modalities, and designed a modality-aware module for more efficient information exchange across different modalities. Mo et al. [2] divided the different modalities into main modality and auxiliary modality, and applied the attention mechanism for feature fusion.

Although the above-mentioned methods make good efforts to utilize multimodal MRI information, they all focus on the extraction and selection of deep semantic features, while ignoring some features with specific meaning and importance to the segmentation problem. In this paper, in addition to the semantic features, we further concentrate on extracting the edge information of multiple types of tumor regions from some relevant modalities including FLAIR and T1ce, which are of great significance to improve segmentation quality, since the edge information is helpful to obtaining more accurate locations and boundaries of tumors. The extracted edge features are merged with the semantic features, aiming to utilize multimodal MRI more effectively and improve the segmentation accuracy accordingly.

3. The Proposed Method

3.1. Overview

The framework of the proposed brain tumor segmentation method is shown in Fig. 2. It is mainly composed of a semantic segmentation module, an edge detection module and a feature fusion module. The semantic module adopts an improved Swin Transformer block to extract deep semantic features from multimodal MRI scans including FLAIR, T1, T1ce and T2. The edge detection module aims to extract edge features by employing a convolutional network as the backbone and designing edge spatial attention blocks (ESABs) for feature enhancement. Considering the modal characteristics of MRI modalities, only FLAIR and T1ce are selected as the input of the edge detection module. The feature fusion module consists of several multi-feature inference blocks (MFIBs), aiming to fuse the semantic features obtained from the semantic segmentation module and the edge features obtained from the edge module at multiple levels. To reconstruct the segmentation result, a successive expanding decoder that is widely adopted in the U-Net-like architectures is employed. For the edge detection task, the result is directly obtained by bilinear interpolation. For the

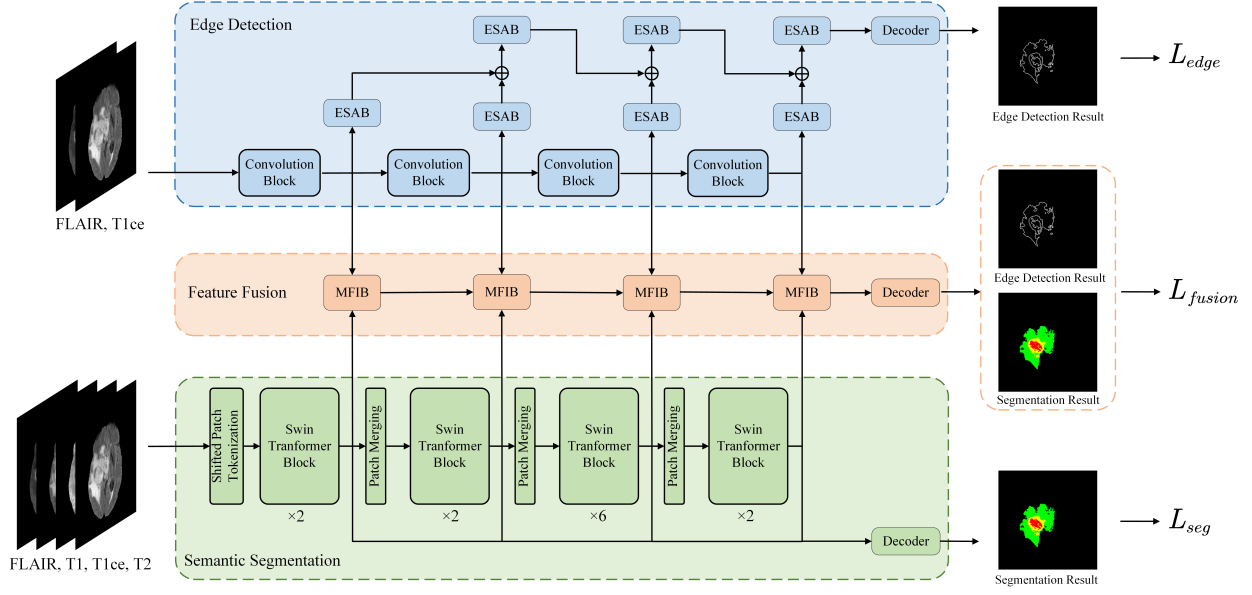


Figure 2: The framework of the proposed brain tumor segmentation method. It mainly consists of three modules: a semantic segmentation module, an edge detection module and a feature fusion module.

feature fusion model, the output includes both an edge detection result and a segmentation result. To train the network, the semantic segmentation module and edge detection module are first trained individually. Then, the output of the feature fusion module is used to supervise the training of the entire model.

3.2. Semantic Segmentation Module

In the semantic segmentation module, we apply the Swin Transformer with an improved patch merging approach to extract semantic features for the segmentation tasks. The Swin Transformer consists of four stages, as shown in Fig. 2. For the last three, the original Swin Transformer blocks with a patch merging step is adopted [11]. For the first one, the steps of patch partition and linear embedding are required prior to the Swin Transformer blocks. Let $X \in \mathbb{R}^{H \times W \times C}$ denote the input, where $H \times W$ represents the size of the input feature map and C represents the number of channels. The input image is first divided into M patches of size $P \times P$, and then each patch is reshaped into a 1D vector $x_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$. Next, these patches are flattened and mapped to D dimension through a trainable linear projection $E \in \mathbb{R}^{(P^2 \cdot C) \times D}$ to obtain the visual token z involving a learnable position variable $E_{pos} \in \mathbb{R}^{(P^2 \cdot C) \times D}$ as

$$z = x_p E + E_{pos}, \quad (1)$$

where z is input to the Transformer Block as an embedding sequence. Since the Transformer directly divides the features, the local information in the patch is difficult to capture, thereby making the Transformer lack the ability of locality inductive bias.

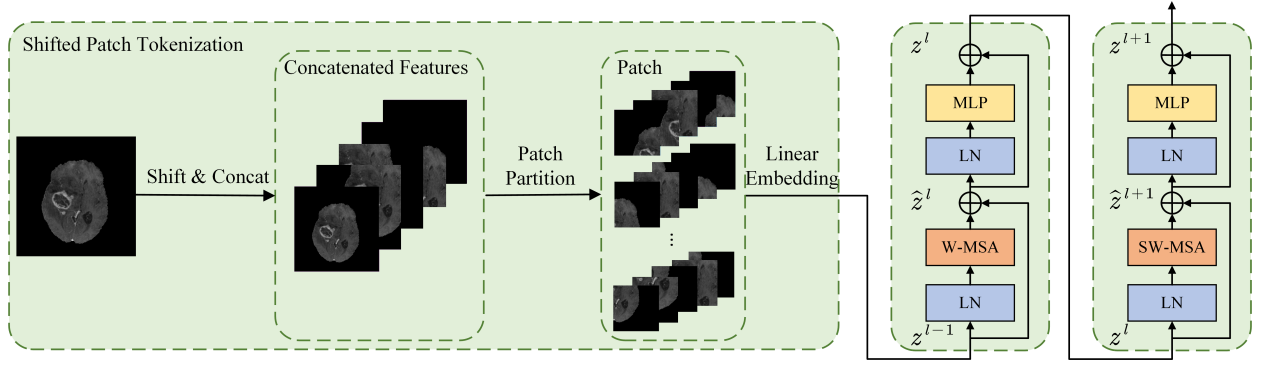


Figure 3: The architecture of a Swin Transformer block with the shifted patch tokenization strategy. Both W-MSA and SW-MSA are multi-head attention, which represent the regular window and the shifted window, respectively.

To address this problem, as shown in Fig. 3, this paper introduces a shifted patch tokenization strategy [12], which can embed more spatial information into the visual token, increasing the locality induction ability of the Transformer to avoid extensive pre-training. Specifically, the input is shifted before patch partition by a patch size from four directions, and then the original input and its shifted versions are spliced. Finally, patch partition and linear embedding are performed.

Each stage in the Swin Transformer consists of L blocks consisting of multi-head self attention (MSA) and multilayer perceptron (MLP). The structure of each block is shown in Fig. 3. The layer normalization (LN) is firstly applied and residual connections are used. The MLP contains two fully connected layers with GELU. The above process can be expressed as

$$\hat{z}^l = \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1}, \quad (2)$$

$$z^l = \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l, \quad (3)$$

$$\hat{z}^{l+1} = \text{SW-MSA}(\text{LN}(z^l)) + z^l, \quad (4)$$

$$z^{l+1} = \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}, \quad (5)$$

where $\hat{z}^l$ and z^l represent the output features of the W-MSA or SW-MSA module and MLP module at the l -th block, where W-MSA and SW-MSA denote window-based multi-head self-attention using regular and shifted window partitioning configurations, respectively. At the end of the Transformer layer, the output z^l goes through a LN layer to obtain the final output z :

$$z = \text{LN}(z^l). \quad (6)$$

To generate a hierarchical representation, the Swin Transformer reduces the number of tokens and

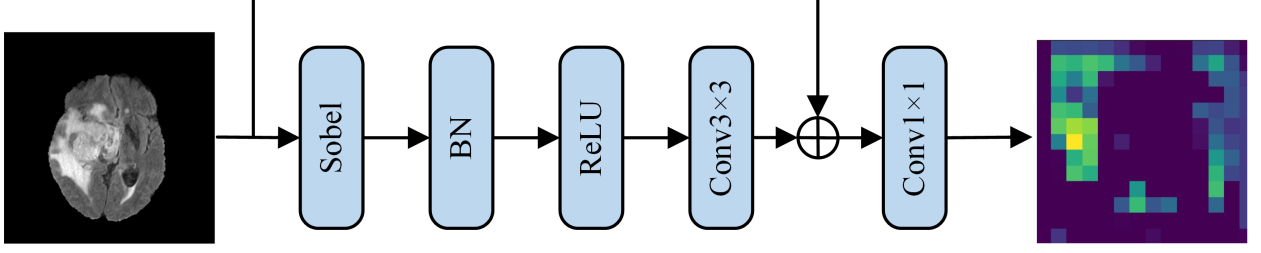


Figure 4: The architecture of the edge spatial attention block (ESAB) designed for edge feature enhancement.

increases the dimensionality through patch merging. In each patch merging step, the features of adjacent 2×2 patches are concatenated and the concatenated features are linearized to increase the dimensionality. Specifically, the following output features are obtained after four stages.

$$\begin{aligned} X_{seg}^1 &\in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times 128}, & X_{seg}^2 &\in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 256} \\ X_{seg}^3 &\in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 512}, & X_{seg}^4 &\in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times 1024} \end{aligned}$$

These features are subsequently input into the feature fusion module and fused with the output features of the edge detection module to achieve better segmentation performance.

3.3. Edge Detection Module

The segmentation performance can be improved by supplementing the edge information of brain tumors. However, the edge features are shallow features. Directly using features of the last convolution block will force the deep network to capture the shallow edge features, thus affecting the extraction performance of the edge features. At the same time, the middle layers can also bring rich convolutional features about edge information [38, 39]. Therefore, it is necessary to utilize all the convolution layers to obtain richer features.

To this end, this paper designs an edge detection module to utilizes the features of multiple convolution layers simultaneously. As shown in the edge detection of Fig. 2, after the image is input into the network, features of different dimensions are extracted through 4 convolution blocks. The convolution block consists of two 3×3 convolutional layers, a regularization layer and a 2×2 max-pooling layer. Specifically, when the image to be detected is input to the edge detection module, the output features obtained after four convolution blocks are given as

$$\begin{aligned} X_{edge}^1 &\in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times 128}, & X_{edge}^2 &\in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 256} \\ X_{edge}^3 &\in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times 512}, & X_{edge}^4 &\in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times 1024} \end{aligned}$$

In order to enhance the edge features, the features of each convolution block are refined by an edge spatial attention block (ESAB). The architecture of the ESAB is shown in Fig. 4. The Sobel convolution is used in the ESAB. The 1×1 convolution reduces the feature volume to a single channel map and obtain

the output feature. Furthermore, the output features of a certain layer are added to the output features of the next layer, leading to a progressive manner to obtain richer edge features. Finally, the output features are interpolated to the original input size to reconstruct edge detection result.

3.4. Feature Fusion Module

The feature fusion module that consists of four multi-feature inference blocks (MFIBs) is designed to fuse semantic features and edge features extracted by the above two modules. In recent years, graph-based applications have become more and more widespread and have been verified to be an effective way of relational reasoning, which makes it a suitable tool to implement multi-feature fusion [40]. In this work, we design the feature fusion module based on graph convolution by referring to a recent work on graph-based global reasoning [41]. The architecture of our MFIB is shown in Fig. 5(a).

The features obtained by the given semantic segmentation module and edge detection module are denoted as $X_{seg} \in \mathbb{R}^{H \times W \times C}$ and $X_{edge} \in \mathbb{R}^{H \times W \times C}$, respectively. To achieve better integration of semantics and features, we map the input features from the spatial domain X to the graph domain $G \in \mathbb{R}^{N \times F}$, where N represents the number of nodes in the graph and F represents the features contained in a node [41]. In this way, pixels with similar features can be aggregated into a node as an anchor to generate a semantic-aware graph feature. The feature fusion process is detailed as below. Let X_{seg} and X_{edge} denote the input semantic and edge features of a given MFIB, respectively. They are mapped to the graph domain to obtain G_{seg} and G_{edge} through two convolutional layers as

$$G_{seg} = v(X_{seg}; W_v) \otimes w(X_{seg}; W_w), \quad (7)$$

$$G_{edge} = v(X_{edge}; W_v) \otimes w(X_{edge}; W_w), \quad (8)$$

where $v(\cdot)$ represents the convolution operations used for graph projection and $w(\cdot)$ represents the convolution operations used for feature dimensionality reduction. W_v and W_w denote the learnable kernels of $v(\cdot)$ and $w(\cdot)$, respectively. The symbol $\otimes$ indicates matrix multiplication. More details of the above projection can be found in [41].

After projection, in order to learn the relationship between the related node features of the semantic graph and edge graph, the graph convolution [42] is adopted to learn edge weights corresponding to the features of each node for reasoning on the fully connected graph. The input of the graph convolution unit G is obtained by adding G_{seg} and G_{edge} as

$$G = G_{seg} + G_{edge}. \quad (9)$$

The architecture of the graph convolution unit is shown in Fig. 5(b), which is implemented by two 1D

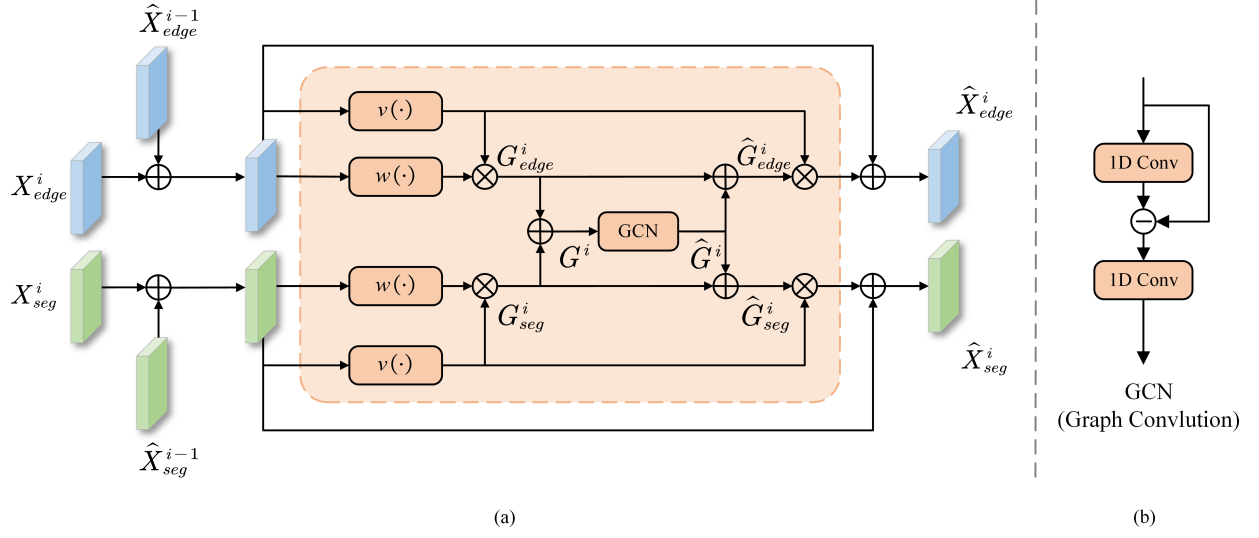


Figure 5: The architecture of the multi-feature inference block (MFIB) for feature fusion. (a) The whole architecture of the i -th MFIB. (b) The specific structure of graph convolution.

convolutions in channel-wise and node-wise directions. As a result, the output can be expressed as

$$\hat{G} = ((I - A_g) G) W_g, \quad (10)$$

where $I \in \mathbb{R}^{N \times N}$ represents the identity matrix, $A_g \in \mathbb{R}^{N \times N}$ represents the adjacency matrix, and W_g represents the update parameter. A_g and W_g are all randomly initialized during training and optimized by gradient descent.

The fused graphs for semantic and edge features are further calculated as

$$\hat{G}_{seg} = G_{seg} + \hat{G}, \quad (11)$$

$$\hat{G}_{edge} = G_{edge} + \hat{G}. \quad (12)$$

Then, we remap $\hat{G}_{seg}$ and $\hat{G}_{edge}$ back to the original spatial domain through the projection operation $v(\cdot)$ obtained in the mapping step to obtain fused features $\hat{X}_{seg}$ and $\hat{X}_{edge}$ as

$$\hat{X}_{seg} = X_{seg} + v(X_{seg}; W_v)^T \otimes \hat{G}_{seg}, \quad (13)$$

$$\hat{X}_{edge} = X_{edge} + v(X_{edge}; W_v)^T \otimes \hat{G}_{edge}. \quad (14)$$

In the proposed method, the input of the first MFIB are exactly the semantic and edge features obtained

at the first stage in the semantic segmentation module and the edge detection module, respectively. For the last three MFIBs, the fused features obtained by the previous block is added into the corresponding original features to generate the input, as shown in Fig. 5(a).

3.5. Loss Function

The proposed segmentation network is trained by three loss functions, L_{seg} , L_{edge} , L_{fusion} . The L_{seg} is the loss of the semantic segmentation module for learning semantic features. The BCEDiceLoss, which is a combination of the binary cross-entropy (BCE) loss and the dice loss [7], is used to define L_{seg} as

$$L_{seg} = \sum \left(0.5 \cdot (-y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})) + \left(1 - \frac{2|y \cap \hat{y}|}{|y| + |\hat{y}|} \right) \right), \quad (15)$$

where y represents GT and $\hat{y}$ represents the prediction result.

The L_{edge} is the loss of the edge detection module for learning edge features. For the edge detection problem, the class imbalance problem is important because most samples are negative. To address this problem, the Ledge is defined by combining the edge loss presented in [38] and the dice loss as

$$L_{edge} = 0.5 \cdot L_{CRF} + L_{Dice}, \quad (16)$$

where the definition of L_{CRF} is given as follows

$$L_i^j = \begin{cases} \alpha \cdot \log(1 - \hat{y}_i^j) & \hat{y}_i = 0 \\ 0 & 0 < \hat{y}_i < \eta, \\ \beta \cdot \log \hat{y}_i^j & other \end{cases} \quad (17)$$

where $\hat{y}_i^j$ represents the predicted value of the i -th pixel of the j -th edge map, and η is a pre-defined threshold. This means that if a pixel marked as positive is by less than η interpreters, it is discarded when the loss is calculated and a positive sample is not considered. β is the number of percentages divided according to negative samples. $\alpha = \lambda \cdot (1 - \beta)$, where λ is a hyperparameter for balancing positive and negative samples.

The L_{fusion} is the loss of the feature fusion module. It is defined as

$$L_{fusion} = L'_{seg} + \gamma \cdot L'_{edge}, \quad (18)$$

where γ is the weight parameter. L'_{seg} and L'_{edge} have the same definitions as L_{seg} and L_{edge} , but using the predictions of the feature fusion module for calculation.

In this paper, we first train the semantic segmentation module and edge detection module using L_{seg} and L_{edge} , respectively. Then, the entire model is further trained using L_{fusion} .

4. Experiments

4.1. Dataset and Implementation Details

In the experiments, the training and testing datasets are all from BraTS2018, BraTS2019 and BraTS2020 benchmarks [43–45]. As an important public dataset for multimodal brain tumor segmentation, BraTS is used in the annual MICCAI brain tumor segmentation challenge and widely adopted in the study of this topic. The dataset is added, deleted or replaced in each year’s competition to enrich its scale. BraTS2018, 2019, and 2020 have 285, 335, and 369 annotated brain tumor samples for model training, respectively. Each case has MRI scans of four different modalities (Flair, T1, T1ce and T2) and are labeled by domain experts. The labels contain four classes: background, NCR/NET, ED and ET. The evaluation is based on three different brain tumor regions: Whole Tumor (WT = NCR/NET + ED + ET), Tumor Core (TC = NCR/NET + ET) and Enhancing Tumor (ET). In this paper, we adopt two most commonly used metrics in medical image segmentation for performance assessment, which are the Dice Score and the %95 Hausdorff Score (HD).

In the preprocessing stage, the size of each scan is $240 \times 240 \times 155$. In this paper, the scans of all modalities are sliced, and the size of each slice is 240×240 . For the semantic segmentation module, all the four modalities are used as the input. For the edge detection module, the input consists of the FLAIR and T1ce modalities. In addition, the popular z-score normalization is performed on the raw data to resolve inconsistencies in image contrast under different modalities.

All the programs were implemented under the PyTorch framework. The training process is conducted on four Tesla P100 GPUs. The optimizer used in the experiments is Adam [46]. The momentum is set to 0.9. The initial learning rate, weight decay and batch size are set to 1e-3, 1e-5, and 16, respectively.

4.2. Parameter Analysis

In Section 3.5, several adjustable parameters such as η , β and γ are involved in the loss functions. The parameters η and β are set to the default values 0.3 and 1.1 according to [?]. In this subsection, we mainly analyze the effect of the parameter γ in Eq. (18) on the segmentation performance obtained by the proposed method. The function of the parameter γ is to balance the semantic segmentation loss and the edge detection loss to ensure both of them have sufficient contributions. In this experiment, we set γ to a set of values including 1, 0.5, 0.1 and 0.05 to study its impact. The BraTS 2018 benchmark is used in this experiment. The corresponding evaluation results are shown in Table 1. It can be seen from the results that the performance on HD tends to be better when γ increases. This is because the edge information is more concerned when γ is larger, leading to higher accuracy in terms of the tumor boundaries, while the metric HD is sensitive to that. For the metric Dice, the proposed method tends to obtain higher performance when a relatively smaller γ is used, which indicates that the semantic features may be more important in

terms of this metric. Based on the above observations, we set γ to 0.1 by default in our method to achieve a good trade-off between the performances on two metrics.

Table 1: The segmentation performance of the proposed method using different values of the parameter γ in Eq. (18).

	WT		TC		ET		Average	
	Dice	HD	Dice	HD	Dice	HD	Dice	HD
$\gamma=1$	90.15	3.364	87.91	4.615	81.37	3.327	86.48	3.769
$\gamma=0.5$	90.85	3.720	88.08	4.569	81.62	3.615	86.85	3.968
$\gamma=0.1$	90.89	3.923	87.96	5.217	81.94	3.440	86.93	4.193
$\gamma=0.05$	90.63	4.419	88.14	5.545	81.41	4.289	86.72	4.289

4.3. Comparison with Other Methods

To verify the superiority of the proposed method for brain tumor segmentation, several state-of-the-art segmentation methods that have been tested on the BraTS2018-2020 benchmarks are used for comparison, which include 2D or 3D CNN-based segmentation methods [3, 13, 14, 27, 28, 47–50], Transformer-based segmentation methods [10, 31] and the methods that focus on multimodal feature fusion [17, 18, 51]. A brief description of these methods is given in Table 2. Since the source code of many existing brain tumor segmentation methods was not released, and to avoid the bias introduced in model re-training, we directly refer to related publications to obtain the evaluation results of the corresponding methods, which is a commonly used manner in the study of brain tumor segmentation. The evaluation results of different methods on BraTS2018, BraTS2019 and BraTS2020 benchmarks are listed in Table 3, Table 4 and Table 5, respectively. The best-performed values are indicated in bold. The corresponding results are visualized for better comparison in Fig. 6 and Fig. 7, which illustrate the performance of different brain tumor segmentation methods on two metrics Dice and HD, respectively. The best-performed method in each case is marked by a star on the corresponding bar.

According to the results reported in the above Tables and Figures, the proposed method achieves more competitive performance when compared with other methods. Specifically, regarding the average Dice score, the proposed method achieves 86.71%, 88.22% and 87.95% on the BraTS2018-2020 benchmarks, which outperforms other reference methods by 0.58% to 7.81%, 0.42% to 8.95%, and 2.89% to 4.93%, respectively. Compared with TransBTS, which jointly uses Transformer and U-Net for semantic segmentation, the proposed method achieves better results in all cases with clear advantages, and the improvement for the tumor core is most significant. Additionally, compared with the latest RFNet method that considers multimodal feature fusion, the proposed method achieves obvious improvement for both tumor core and enhancing tumor regions.

Fig. 8 shows the visual effect comparison of the brain tumor segmentation results obtained by different

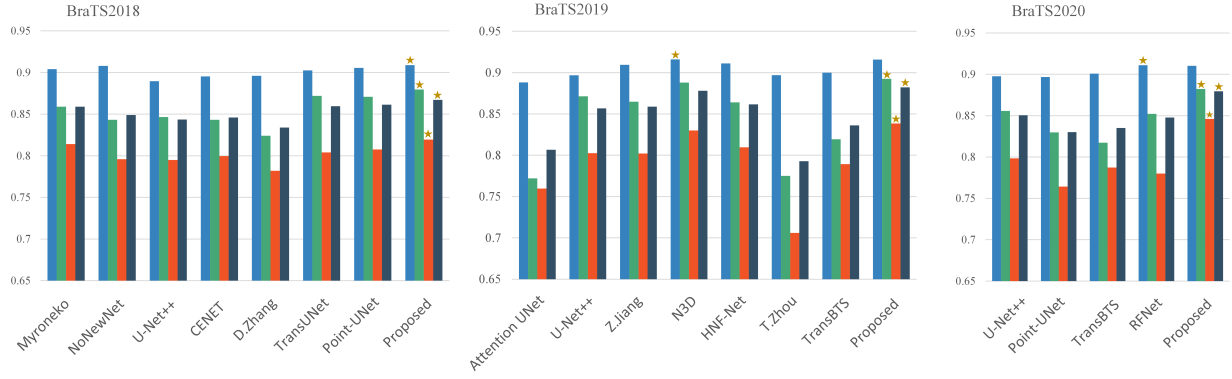


Figure 6: Performance comparison of different brain tumor segmentation methods on the metric Dice. The best-performed method in each case is marked by a star.

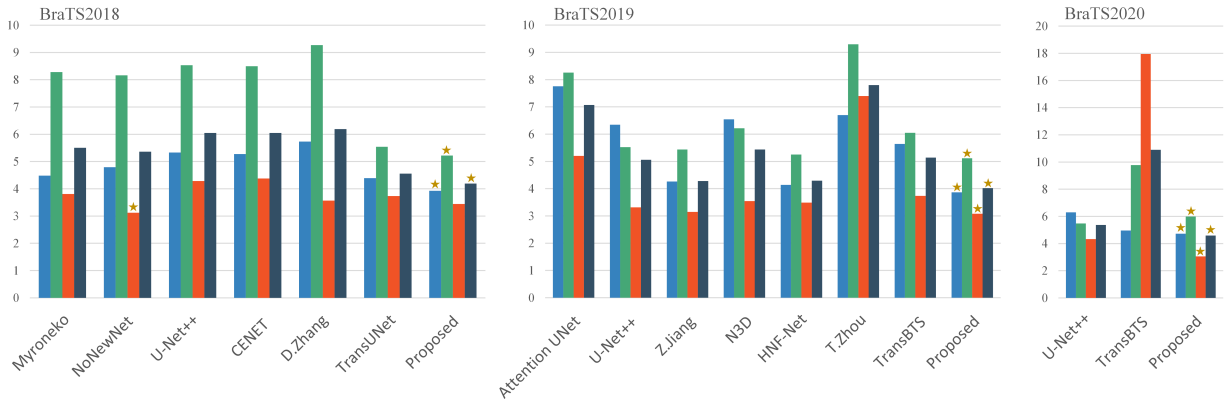


Figure 7: Performance comparison of different brain tumor segmentation methods on the metric HD. The best-performed method in each case is marked by a star.

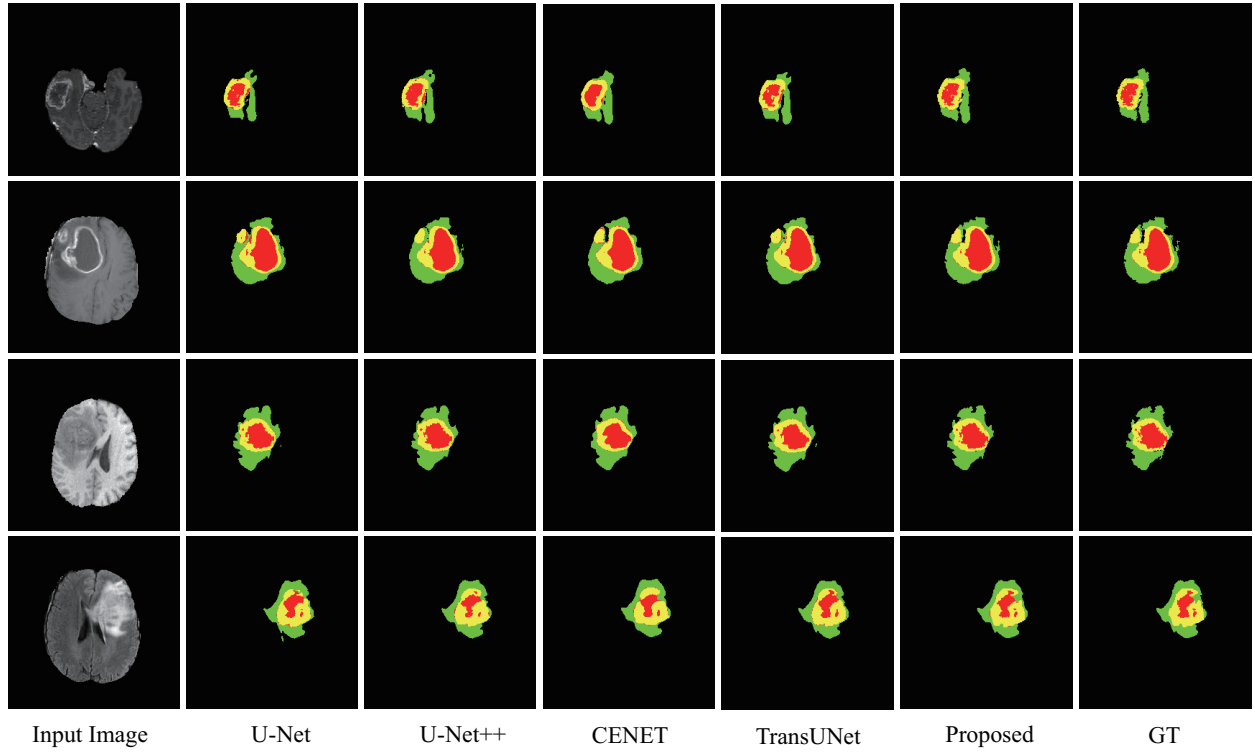


Figure 8: Visual effect comparison of brain tumor segmentation results obtained by different methods. The green, yellow and red indicate ED, ET and NCR/NET regions, respectively.

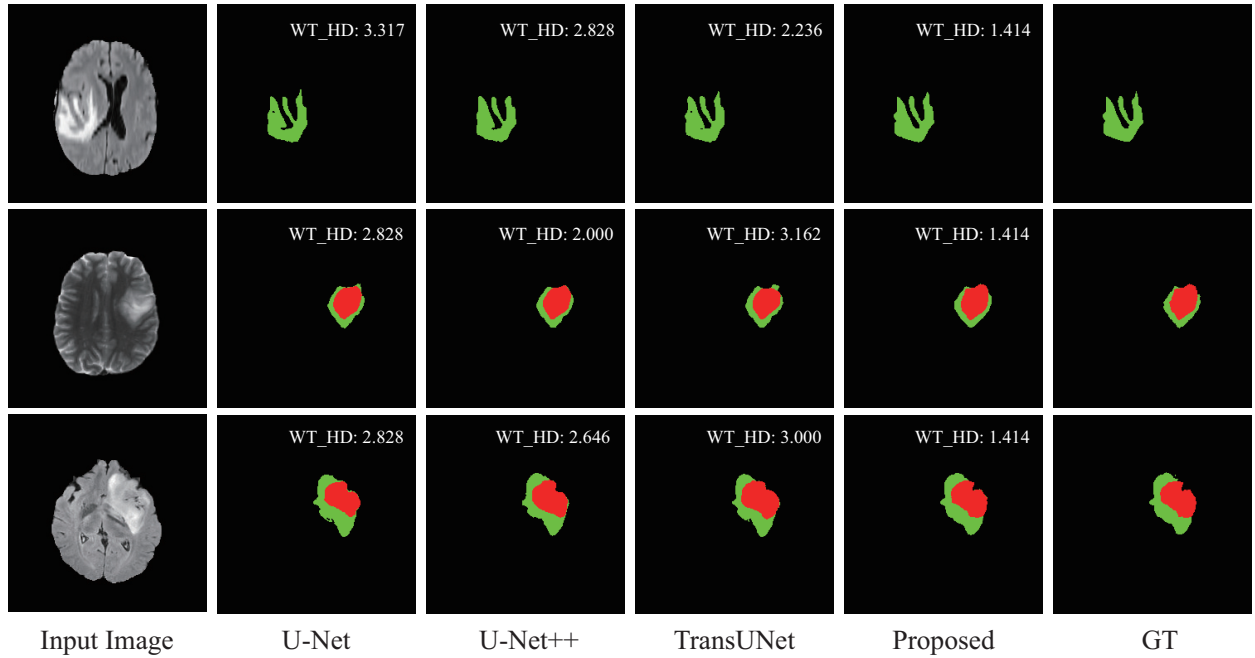


Figure 9: Performance comparison of different segmentation methods in terms of tumor boundary accuracy.

Table 2: A brief description of the methods used for performance comparison in our experiments.

Method	Brief description
Myronenko[3]	Proposes a U-Net-based segmentation framework by adding a VAE branch to regularize feature learning.
NoNewNet[13]	Designs an improved U-Net architecture for segmentation.
Attention Unet[27]	Proposes an attention mechanism based on U-Net for segmentation.
U-Net++[28]	Adds a series of dense skip connections to the U-Net for segmentation.
N3D[14]	Proposes a 3D U-Net for brain tumor segmentation.
Z. Jiang[49]	Proposes an end-to-end cascading U-Net architecture for segmentation.
CENET[47]	Designs a context extractor to generate more advanced semantic feature maps.
HNF-Net[50]	Proposes a 3D high-resolution and non-local feature network for segmentation.
T. Zhou[18]	Designs four independent encoding paths to extract features from four modalities and then fuse them.
D. Zhang[17]	Proposes a task-structured brain tumor segmentation network by considering multimodal fusion.
TransBTS[10]	Proposes an encoder-decoder structure consisting of Transformer and U-Net for segmentation.
RFNet[51]	Proposes a region-aware fusion network that exploits different combinations of multimodal data.
TransUnet[31]	Proposes a universal segmentation framework by combining Transformer and U-Net.
Point-UNet[48]	Designs a U-Net to perform a fine-class segmentation of the input point cloud.

Table 3: Objective evaluation results of different brain tumor segmentation methods on the BraTS2018 benchmark.

	WT		TC		ET		Average	
	Dice	HD	Dice	HD	Dice	HD	Dice	HD
Myronenko[3]	90.40	4.483	85.90	8.278	81.40	3.805	85.90	5.500
NoNewNet[13]	90.80	4.790	84.32	8.160	79.59	3.120	84.90	5.357
U-Net++[28]	88.96	5.327	84.65	8.535	79.49	4.285	84.36	6.049
CENET[47]	89.53	5.271	84.31	8.493	79.95	4.379	84.60	6.193
D. Zhang[17]	89.60	5.733	82.40	9.270	78.20	3.567	83.40	6.190
TransUnet[31]	90.25	4.390	87.19	5.539	80.41	3.731	85.95	4.553
Point-UNet[48]	90.55	-	87.09	-	80.76	-	86.13	6.010
Proposed	90.89	3.923	87.96	5.217	81.94	3.440	86.93	4.193

methods. By referring to the ground truth (GT), the proposed method achieves more accurate segmentation results and especially for the tumor edges than other methods, which demonstrates the effectiveness of the edge features extracted for segmentation.

Fig. 9 illustrates an example to compare the performance of different segmentation methods in terms of tumor boundary accuracy. As mentioned above, the metric HD is more sensitive to the boundary shape, so the corresponding HD scores of whole tumor are provided as well. Among all the methods, the proposed one obtains the best results in terms of both the HD scores and the visual effect. These results further show

Table 4: Objective evaluation results of different brain tumor segmentation methods on the BraTS2019 benchmark.

	WT		TC		ET		Average	
	Dice	HD	Dice	HD	Dice	HD	Dice	HD
Attention Unet[27]	88.81	7.756	77.20	8.258	75.96	5.202	80.66	7.072
U-Net++[28]	89.67	6.345	87.13	5.521	80.25	3.313	85.68	5.060
Z. Jiang[49]	90.94	4.263	86.47	5.439	80.21	3.146	85.87	4.283
N3D[14]	91.60	6.547	88.80	6.219	83.00	3.543	87.80	5.436
HNF-Net[50]	91.11	4.136	86.40	5.250	80.96	3.490	86.16	4.292
T. Zhou[18]	89.70	6.700	77.50	9.300	70.60	7.400	79.27	7.800
TransBTS[10]	90.00	5.644	81.94	6.049	78.93	3.736	83.62	5.143
Proposed	91.58	3.866	89.24	5.118	83.84	3.080	88.22	4.021

Table 5: Objective evaluation results of different brain tumor segmentation methods on the BraTS2020 benchmark.

BraTS2020	WT		TC		ET		Average	
	Dice	HD	Dice	HD	Dice	HD	Dice	HD
U-Net++[28]	89.77	6.299	85.57	5.483	79.83	4.328	85.06	5.370
Point-UNet[48]	89.67	-	82.97	-	76.43	-	83.02	8.260
TransBTS[10]	90.09	4.964	81.73	9.769	78.73	17.947	83.52	10.893
RFNet[51]	91.11	-	85.21	-	78.00	-	84.77	-
Proposed	91.03	4.719	88.22	5.985	84.61	3.051	87.95	4.585

that edge features can benefit the brain tumor segmentation task.

4.4. Ablation Study

To further verify the effectiveness of the main components including the shifted patch tokenization strategy, the edge detection module and the MFIB in the feature fusion module that are designed in our method, a ablation study is conducted in this subsection.

Table 6: Objective evaluation results for the ablation study on the BraTS2018 benchmark.

	WT		TC		ET		Average	
	Dice	HD	Dice	HD	Dice	HD	Dice	HD
SwinTrans	88.97	6.276	85.72	6.563	80.31	4.364	85.00	5.734
SwinTrans+SPD	89.00	5.720	85.95	6.453	80.62	4.338	85.19	5.504
SwinTrans+SPD+ED	89.93	4.259	87.24	5.398	81.19	3.728	85.95	4.462
Proposed	90.89	3.923	87.96	5.217	81.94	3.440	86.93	4.193

In this experiment, we use the standard Swin Transformer [11] as the baseline model, and then adds different components one by one to validate their effect. Specifically, we mainly compare the performance of the following four models:

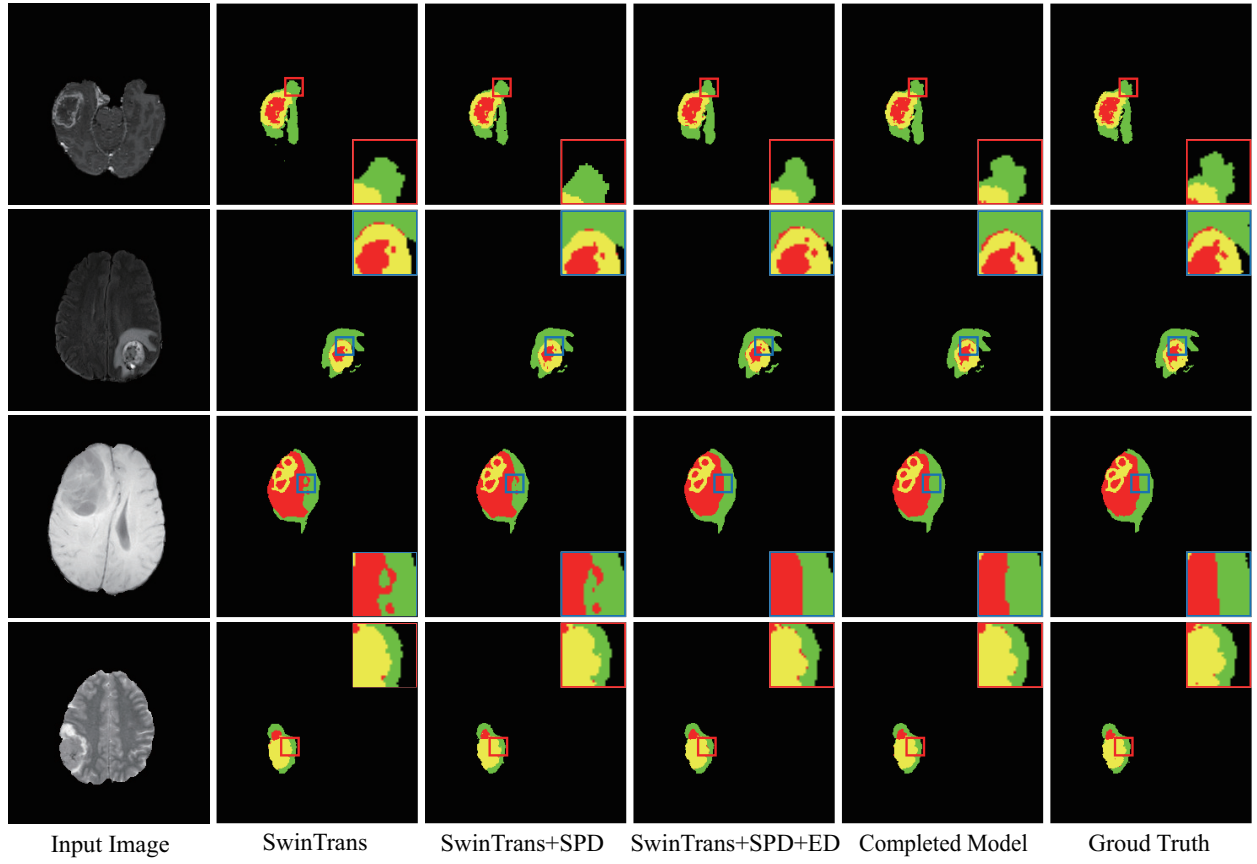


Figure 10: Visual effect comparison of segmentation results obtained by different models in the ablation study.

- **SwinTrans**: Just using the Swin Transformer for brain tumor segmentation via a well pre-trained model. This is the baseline model.
- **SwinTrans+SPD**: Introducing the shifted patch tokenization strategy into the Swin Transformer for segmentation without using pre-training.
- **SwinTrans+SPD+ED**: Further adding the edge detection module based on the above model. This model has similar framework to the proposed one, but it just simply concatenate the semantic and edge features for fusion, instead of using the proposed MFIB.
- **Completed Model**: The complete model (i.e., SwinTrans+SPD+ED+MFIB) proposed in this paper.

Therefore, the comparison between SwinTrans and SwinTrans+SPD is used to demonstrate the effectiveness of the shifted patch tokenization strategy adopted in the Swin Transformer. The comparison between SwinTrans+SPD and SwinTrans+SPD+ED can validate the effect of the edge detection module (please kindly note that the ED module cannot be individually used without the semantic segmentation module). The comparison between SwinTrans+SP+ED and Completed Model is used to show the effectiveness of the designed MFIB for feature fusion.

Table 6 lists the objective performance of different models. We can see that the each of the above components leads to some improvements of the segmentation results. Among them, the effect of adding edge detection module and using MFIB for feature fusion is more obvious.

The visual effect comparison of segmentation results obtained by different models in the ablation study is shown in Fig. 10. Some interesting observations include: 1) After adding the shifted patch, partial unnecessary disturbance noise is eliminated. 2) After adding the edge detection module, the segmentation accuracy of tumor edges is obviously higher when compared with the GT. 3) After adding the MFIB for feature fusion, the tumor edges are visually more natural than the simple concatenating manner.

5. Conclusion

This paper proposes a novel deep learning-based brain tumor segmentation method by jointly utilizing deep semantics and edge information in multimodal MRI. To achieve this target, three functional modules are designed. Specifically, we present a semantic segmentation module based on an improved Swin Transformer by introducing the shifted patch tokenization strategy for better training. In addition, a CNN-based edge detection module is designed to extract edge features from the input MRI scans. Finally, we present a feature fusion module by designing a multi-feature inference block based on graph convolution to fuse the deep semantic edges and specific edge features. Experimental results demonstrate the effectiveness of the key components designed in our method. Moreover, the proposed method achieves better performance when compared with some state-of-the-art methods on the BraTS benchmarks. Future work may focus

on exploring the feasibility of some other specific features for brain tumor segmentation and extending the proposed approach to other semantic segmentation problems.

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